



Stepwise Self-Assembly and Dynamic Exchange of Supramolecular Nanocages Based on Terpyridine Building Blocks

Zhe Zhang, Heng Wang, Junjuan Shi, Yaping Xu, Lei Wang, Sammy Shihadeh, Fu-Jie Zhao, Xin-Qi Hao, Pingshan Wang, Changlin Liu, Ming Wang,* and Xiaopeng Li*

Coordination-driven self-assembly as a powerful bottom-up approach has been extensively used to construct multifarious supramolecular architectures with increasing complexity and functionality. Due to the unique cavity structures and precisely controllable dimensions, 3D supramolecules display unprecedented properties and functions in catalysis, sensing, gas storage, and smart materials. Herein, we have built two 3D nanocages with different sizes by changing the length of the organic ligand arms. The structures were characterized by 1D and 2D NMR spectroscopy, electrospray ionization-mass spectrometry (ESI-MS), traveling wave ion mobility-mass spectrometry (TWIM-MS), gradient tandem-mass spectrometry (gMS²), and transmission electron microscopy (TEM). Furthermore, the intermolecular dynamic exchange of two 3D nanocages was conducted to construct a series of hybrid 3D structures as evidenced by mass spectrometry.

Self-assembly as a powerful bottom-up approach widely exists in the formation of complicated functional structures.^[1] Synthetic supramolecular chemistry focused on the construction of specific and functionalized supramolecular architectures through self-assembly during the past few decades. Coordination-driven self-assembly is one of the most useful methods for preparing multifarious supramolecular structures,^[2] such as 1D helicates, 2D macrocycles, knots, links, 3D cages, and capsules.^[1–13] 3D supramolecules with well-defined cavities are attractive due to their ability of encapsulating guest molecules. As a result, they have broad applications in catalysis,^[14] drug delivery,^[15] molecule recognition,^[16] stabi-

lization of (air)-sensitive molecules,^[7,17] etc.

Terpyridine (tpy)-based building blocks that coordinate with transition metal ions (i.e., <tpy-M(II)-tpy>) have been applied for constructing supramolecules.^[18] Numerous tpy-based metallo-supramolecules, including triangle,^[19] hexagons,^[20] fractals,^[21] spoked wheels,^[22] and nested concentric ring^[23] have been reported. Despite several elegant reports,^[24,25] the synthesis of tpy-based 3D nanocages is still challenging in terms of diversity and complexity.

Recently, we reported two distinct 2D supramolecular snowflakes,^[26] which were obtained by self-assembling metal-organic ligand containing a <tpy-Ru(II)-tpy> motif with a hexatopic ligand and Zn(II) ions. Interestingly, we found that the metal-organic ligand itself can coordinate with Zn(II) ions to form 3D nanocage, which is similar to Stang's truncated tetrahedron^[27] assembled by pyridine ligand. Herein, we have built two 3D nanocages, with different sizes by changing the length of the metal-organic ligand arms. The further dynamic ligand exchange between two preassembled nanocages generated as a series of hybrid supramolecular nanocages, suggesting the remarkable tolerance of such structures with different edge length.

With structural analysis and molecular modeling, we speculated that three-armed tpy-based organic ligand might assemble with metal components to build a truncated tetrahedron-shaped 3D nanocage structure (**Scheme 1**). The metal-organic ligand **L1** containing three free terpyridines was designed with **V** and **Y** components connected by Ru(II). Such building block was

Dr. Z. Zhang, Prof. P. Wang
Key Laboratory for Water Quality and Conservation of the Pearl River Delta
Ministry of Education
Environmental Research at Great Bay
Guangzhou University
Guangzhou 510006, P. R. China
Prof. X. Li, Dr. Z. Zhang, Dr. H. Wang, L. Wang, S. Shihadeh
Department of Chemistry
University of South Florida
Tampa 33620, USA
E-mail: xiaopengli1@usf.edu

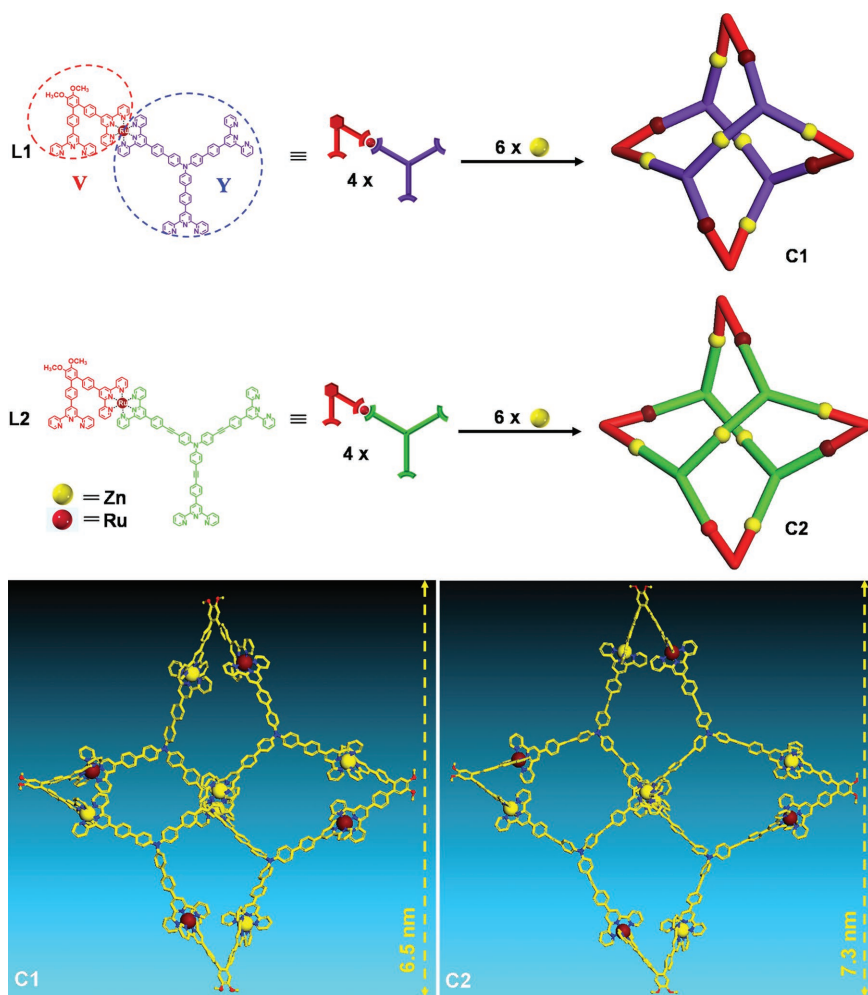
Prof. M. Wang, J. Shi, Y. Xu
State Key Laboratory of Supramolecular Structure and Materials
College of Chemistry
Jilin University
Changchun 130012, P. R. China
E-mail: mingwang358@jlu.edu.cn

Prof. C. Liu, Dr. Z. Zhang
Key Laboratory of Pesticide and Chemical Biology
Ministry of Education
School of Chemistry
Central China Normal University
Wuhan 430079, P. R. China

Prof. X.-Q. Hao, F.-J. Zhao
College of Chemistry and Molecular Engineering
Zhengzhou University
Zhengzhou 450001, P. R. China

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/marc.201800404>.

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Scheme 1. Self-assembly of 3D supramolecular nanocages and representative energy-minimized structures of **C1** and **C2**.

synthesized by performing Suzuki coupling reaction on Ru(II)-tpy complex to avoid the by-products via self-sorting as shown in Scheme S1, Supporting Information. **C1** was obtained in high yield (94.3%) by self-assembling **L1** with $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$.

The ^1H NMR spectrum of ligand **L1** (Figure 1a) showed four distinguishable sets of tpy-phenyl resonances peaks (A–D, D, and E sets are chemical equivalent). However, after coordination with Zn(II), the ^1H NMR spectrum of **C1** became further complicated with five sets of tpy-phenyl signals (labeled as A–E). Two sets of peaks at 9.18 and ≈ 9.08 ppm for the $\text{tpyH}^{3,5'}$ protons of <tpy-Ru(II)-tpy> moieties showed negligible shifts. After coordinating with Zn(II) ion, all the free $\text{tpyH}^{3,5'}$ protons of **L1** were significantly shifted to downfield due to the decrease of electron density upon complexation.^[17] The expected upfield shift of doublet from 8.7 to 7.9 ppm (≈ 0.8 ppm) was attributed to the movement of $\text{tpyH}^{6,6''}$ protons after coordination. All of the assignments were readily confirmed by the 2D-COSY and 2D-NOESY spectra (Figures S23–S29, Supporting Information). 2D diffusion-ordered NMR spectroscopy (DOSY) (Figure 1c) showed a single band near a diffusion coefficient of $1.85 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ (log D values as -9.73). Using the

Stocks–Einstein equation, the experimental radius, that is, the semimajor axis radius, of **C1** was 3.2 nm, was comparable with outer radius measured from the molecular modeling of **C1** (≈ 3.3 nm, Scheme 1).

In addition, ESI-MS coupled with TWIM-MS and gMS² was applied to validate the proposed structure. ESI-MS showed a series of peaks with continuous charges from 7+ to 15+ due to successive loss of counterion. After deconvolution, the obtained molecular weight at 11365 Da agreed well with the proposed molecular composition $[(\text{C}_{131}\text{H}_{90}\text{N}_{16}\text{O}_2)_4\text{Zn}_6\text{Ru}_4(\text{PF}_6^-)_{20}]$ (Figure 2a). The experimental isotope patterns of each charge state was consistent with the simulated isotopic distribution (Figure S1, Supporting Information). TWIM-MS showed a series of charge states with narrow drift time distribution ranging from 10+ to 17+, excluding the formation of any isomers or oligomers (Figure 2b). In order to examine the stability of the **C1**, gMS² experiment was performed on the 13+ ions at m/z 730.1 by collision-induced dissociation with collision energies ranging from 5 to 20 V (Figure S3, Supporting Information). The complex ions were completely dissociated when the voltage reached 20 V, which was converted to a center-of-mass collision energy of 0.08 eV. This result was used in the comparison with **C2**.

TEM was utilized to further confirm both shapes and sizes of individual **C1** molecule by deposition of the dilute MeCN solution on carbon-coated grids. Individual circular structure with high

contrast was observed for **C1**. The measured size was consistent with the theoretical diameter of 6.5 nm from molecular modeling (Figure 3a).

On the basis of **C1**, we stepped forward to construct larger version of supramolecular nanocage **C2** by increasing the arm length of **Y**. **L2** was prepared by introducing alkyne bond in a triphenylamine moiety, then assembled with Zn(II) to produce **C2**. We expected that this rigid and stable system with truncated tetrahedron shaped geometry was able to accommodate the variation of ligand arm length but without losing the entire structure integrity benefitting from the energy favorable features of the structure.

Figure 1b showed the ^1H NMR spectra of ligand **L2** and complex **C2**. After assembly characteristic downshift and upshift of (A, D, E)- $\text{tpyH}^{3,5'}$ and $\text{tpyH}^{6,6''}$ protons together indicated the formation of the designed structure. DOSY showed a single band near a diffusion coefficient of $1.55 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ (log D values as -9.81) with larger conducted semimajor axle radius (3.8 nm) (Figure 1d). TEM imaging of **C2** also displayed clear images of circular structure with narrow dispersity of diameter around 7.5 nm (Figure 3b).

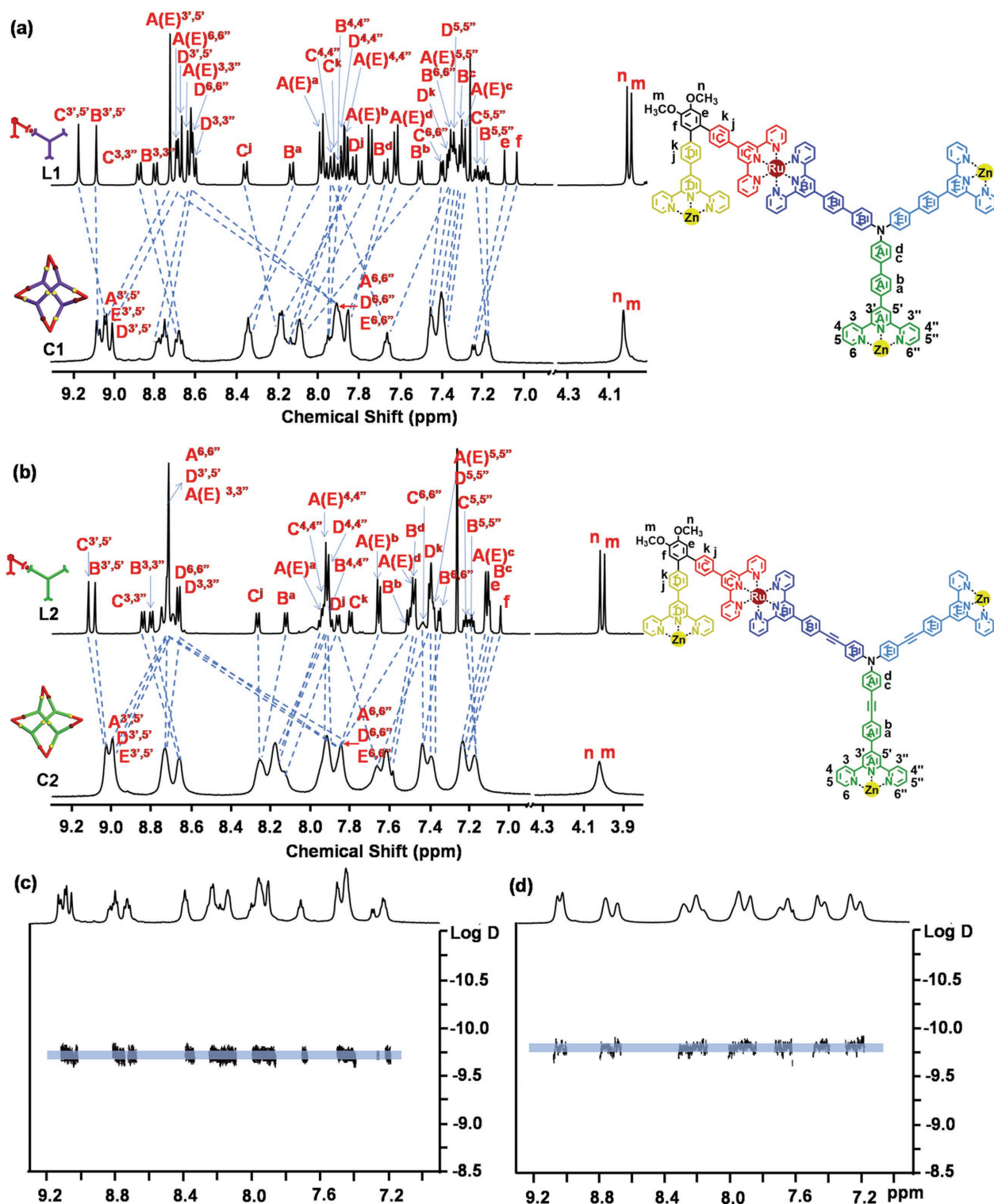


Figure 1. ^1H NMR spectra. a) L1 and C1; b) L2 and C2; c) DOSY of C1; d) DOSY of C2 (CDCl_3 for ligands and CD_3CN for supramolecules).

ESI-MS result of C2 (Figure 2c) displayed a series of peaks with continuous charge states from 8+ to 17+. The measured molecular weight of C2 was 11653 Da, supporting its composition $[(\text{C}_{137}\text{H}_{90}\text{N}_{16}\text{O}_2)_4\text{Zn}_6\text{Ru}_4(\text{PF}_6^-)_{20}]$. TWIM-MS displays one predominant series of signals with narrow distribution

(Figure 2d), suggesting the absence of any kind of isomer. The 13+ ions of C2 were completely dissociated at 21 V in gMS^2 (Figure S4, Supporting Information), which was converted to a center-of-mass collision energy of 0.08 eV, same as that of C1. Therefore, C1 and C2 possess similar stability.

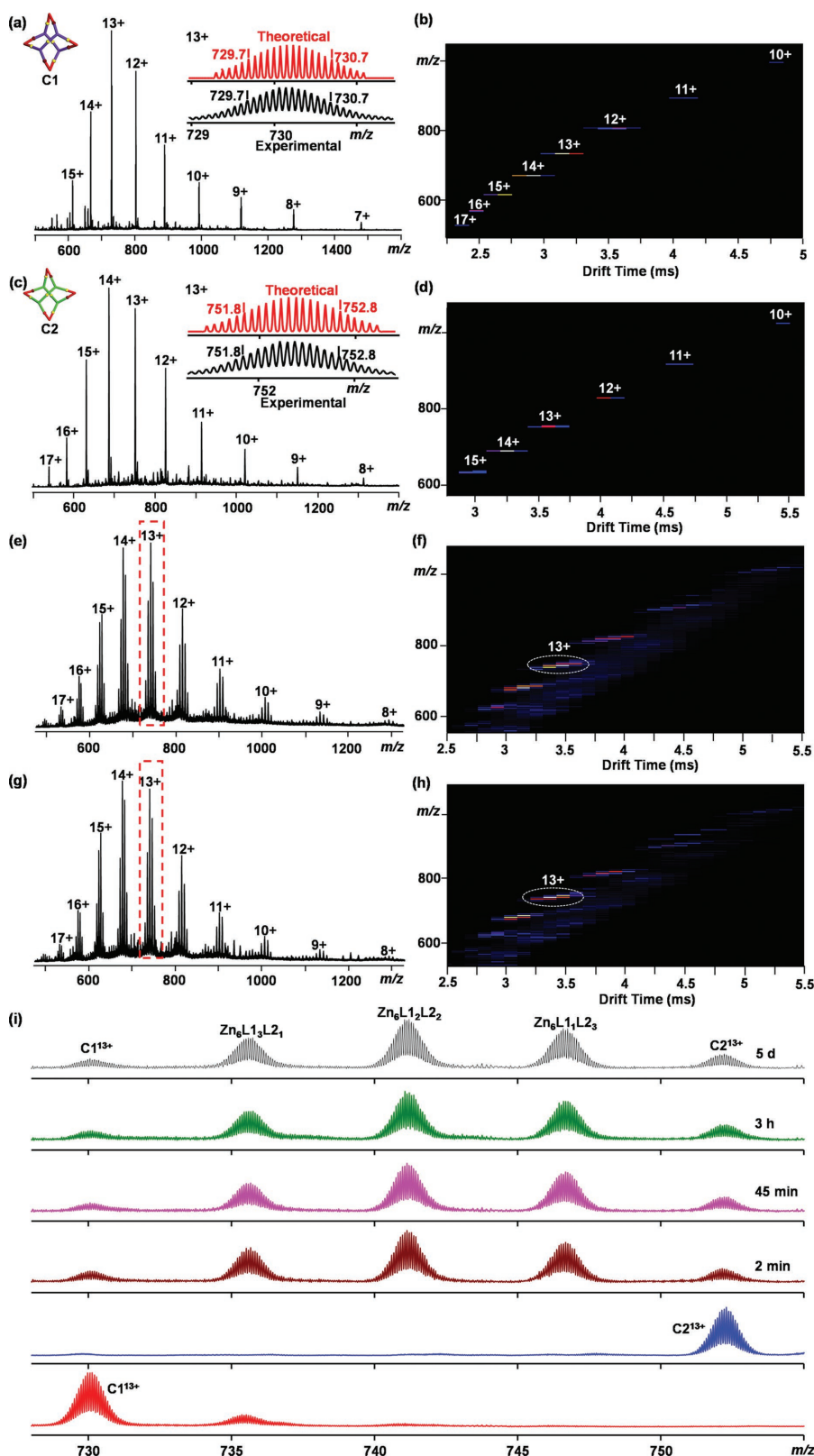


Figure 2. a,c) ESI-MS and b,d) TWIM-MS plot (m/z vs drift time) of **C1** and **C2**. e–i) Dynamic ligand exchange of preassembled **C1** and **C2**. e) ESI-MS and f) TWIM-MS plot of preassembled **C1** and **C2** at 2 min. g) ESI-MS and h) TWIM-MS plot of preassembled **C1** and **C2** at 5 d. i) Time-dependent spectra of dynamic exchange of **C1** and **C2** at 2 min, 45 min, 3 h, 5 d for expanded region of 13+ signals.

Despite the difference of the ligands, **C1** and **C2** could be assembled with similar structure and stability. The next question we raised is whether we were able to assemble hybrid supramolecular nanocages containing both **L1** and **L2** or not. In our previous study,^[26] ligand exchange between different snowflakes with various outer rims has been fully explored. In this work, we are curious about the intermolecular communication feature between **C1** and **C2**, after ascending 2D snowflake structure into 3D nanocage.

We started the study by mixing preassembled **C1** and **C2** (CH_3CN , 1 mg mL^{-1}) solutions with 1:1 ratio at room temperature. All the experimental process was monitored by ESI-MS and TWIM-MS (Figure 2e–i). The dynamic ligand exchange started and almost finished within 2 min after the mixing. In addition to the original peaks of **C1** and **C2**, for example **C1**¹³⁺ and **C2**¹³⁺, three new peaks appeared, corresponding to all the statistical hybrid 3D nanocages. No significant change was observed by extending the monitoring time (Figure 2i). The result showed that the dynamic ligand exchange between **C1** and **C2** was substantially faster than that between two snowflake supramolecules, as the dynamic exchange between snowflakes reached equilibrium around 5 days. Such difference suggests that the energy barrier of the ligand communication between nanocages is much lower than that between two snowflakes. In the spectrum of TWIM-MS (Figure 3f,h), all the hybrid signals drifted out with very similar times, indicating that they should have the similar 3D nanocage structure as **C1** and **C2**.

Considering the tpy and triphenylamine have a wide range of applications in optoelectronics, we studied the photophysical and electrochemical properties of the complexes. In the absorption spectra (Figure S5, Supporting Information), **C1** and **C2** showed characteristic absorption bands peaked at 380 and 500 nm, which corresponds to the intramolecular charge transfer (ICT) transition from triphenylamine moiety to the tpy-metal complex moieties and metal-to-ligand charge transfer (MLCT) transitions, respectively.^[28] Oxidation and reduction

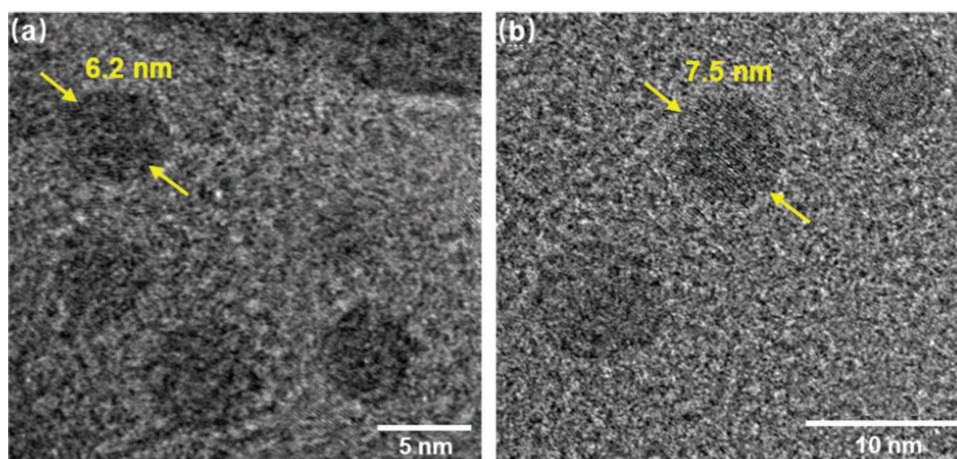


Figure 3. TEM images of a) **C1** and b) **C2**.

properties of these two complexes were determined through cyclic voltammetry (CV) with results displayed in Figure S6, Supporting Information. Two tpy-ligand-centered redox couples were observed between -2 and -0.5 V.

In summary, by taking advantage of the diversity of the coordination ability between terpyridine with transition metal ions, two supramolecular nanocages were assembled successfully by stepwise approach. The structures were characterized by 1D and 2D NMR, ESI-MS, TWIM-MS, gMS², TEM, along with molecular modeling. Furthermore, the dynamic intermolecular exchange behavior between two 3D nanocages was investigated by ESI-MS and TWIM-MS. It showed that such versatile supramolecular structures were able to tolerate different types of ligands with various combinations and form hybrid supramolecular nanocages. The rapid communication between supramolecules via ligand exchange further increased the complexity of the system. Electrochemical and photophysical properties of **C1** and **C2** were also investigated for future potential applications, such as host–guest, supramolecular catalysis, gas storage, etc.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

3D nanocages, self-assembly, supramolecules, terpyridine

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- [1] a) J.-M. Lehn, *Science* **2002**, 295, 2400; b) A. V. Davis, R. M. Yeh, K. N. Raymond, *Proc. Nat. Acad. Sci. U. S. A.* **2002**, 99, 4793; c) F. Wurthner, C.-C. You, C. R. Saha-Moller, *Chem. Soc. Rev.* **2004**, 33, 133.
- [2] a) T. R. Cook, Y.-R. Zheng, P. J. Stang, *Chem. Rev.* **2013**, 113, 734; b) R. Chakrabarty, P. S. Mukherjee, P. J. Stang, *Chem. Rev.* **2011**, 111, 6810; c) S. Mukherjee, P. S. Mukherjee, *Chem. Commun.* **2014**, 50, 2239.
- [3] a) S. R. Seidel, P. J. Stang, *Acc. Chem. Res.* **2002**, 35, 972; b) M. Fujita, M. Tomimaga, A. Hori, B. Therrien, *Acc. Chem. Res.* **2005**, 38, 369.
- [4] a) B. Olenyuk, J. A. Whiteford, A. Fechtenkötter, P. J. Stang, *Nature* **1999**, 398, 796.
- [5] D. Fujita, Y. Ueda, S. Sato, N. Mizuno, T. Kumasaka, M. Fujita, *Nature* **2016**, 540, 563.
- [6] C. Zhao, Q.-F. Sun, W. M. Hart-Cooper, A. G. DiPasquale, F. D. Toste, R. G. Bergman, K. N. Raymond, *J. Am. Chem. Soc.* **2013**, 135, 18802.
- [7] a) P. Mal, B. Breiner, K. Rissanen, J. R. Nitschke, *Science* **2009**, 324, 1697; b) I. A. Riddell, Y. R. Hristova, J. K. Clegg, C. S. Wood, B. Breiner, J. R. Nitschke, *J. Am. Chem. Soc.* **2013**, 135, 2723.
- [8] J. J. Danon, A. Krüger, D. A. Leigh, J.-F. Lemonnier, A. J. Stephens, I. J. Vitorica-Yrezabal, S. L. Woltering, *Science* **2017**, 355, 159.
- [9] D. A. Fowler, A. S. Rathnayake, S. Kennedy, H. Kumari, C. M. Beavers, S. J. Teat, J. L. Atwood, *J. Am. Chem. Soc.* **2013**, 135, 12184.
- [10] S. Hiraoka, Y. Yamauchi, R. Arakane, M. Shionoya, *J. Am. Chem. Soc.* **2009**, 131, 11646.
- [11] C. G. H., K. Wataru, T. Shohei, S. Motoo, S. Mitsuhiro, *Angew. Chem. Int. Ed.* **2012**, 51, 2606.
- [12] J.-R. Li, A. A. Yakovenko, W. Lu, D. J. Timmons, W. Zhuang, D. Yuan, H.-C. Zhou, *J. Am. Chem. Soc.* **2010**, 132, 17599.
- [13] a) I. A. Bhat, D. Samanta, P. S. Mukherjee, *J. Am. Chem. Soc.* **2015**, 137, 9497; b) B. Song, Z. Zhang, K. Wang, C.-H. Hsu, O. Bolarinwa, J. Wang, Y. Li, G.-Q. Yin, E. Rivera, H.-B. Yang, C. Liu, B. Xu, X. Li, *Angew. Chem. Int. Ed.* **2017**, 56, 5258.



- [14] V. Marcos, A. J. Stephens, J. Jaramillo-Garcia, A. L. Nussbaumer, S. L. Woltering, A. Valero, J.-F. Lemonnier, I. J. Vitorica-Yrezabal, D. A. Leigh, *Science* **2016**, 352, 1555.
- [15] S. K. Samanta, D. Moncelet, V. Briken, L. Isaacs, *J. Am. Chem. Soc.* **2016**, 138, 14488.
- [16] S. J. Barrow, S. Kaser, M. J. Rowland, J. del Barrio, O. A. Scherman, *Chem. Rev.* **2015**, 115, 12320.
- [17] P. Mal, B. Breiner, K. Rissanen, J. R. Nitschke, *Science* **2009**, 324, 1697.
- [18] S. Chakraborty, G. R. Newkome, *Chem. Soc. Rev.* **2018**, <https://doi.org/10.1039/c8cs00030a>.
- [19] A. Schultz, Y. Cao, M. Huang, S. Z. D. Cheng, X. Li, C. N. Moorefield, C. Wesdemiotis, G. R. Newkome, *Dalton Trans.* **2012**, 41, 11573.
- [20] G. R. Newkome, T. J. Cho, C. N. Moorefield, G. R. Baker, R. Cush, P. S. Russo, *Angew. Chem. Int. Ed.* **1999**, 38, 3717.
- [21] a) G. R. Newkome, P. Wang, C. N. Moorefield, T. J. Cho, P. P. Mohapatra, S. Li, S.-H. Hwang, O. Lukyanova, L. Echegoyen, J. A. Palagallo, V. Iancu, S.-W. Hla, *Science* **2006**, 312, 1782; b) Z. Jiang, Y. Li, M. Wang, D. Liu, J. Yuan, M. Chen, J. Wang, G. R. Newkome, W. Sun, X. Li, P. Wang, *Angew. Chem. Int. Ed.* **2017**, 56, 11450;
- [22] X. Lu, X. Li, Y. Cao, A. Schultz, J. L. Wang, C. N. Moorefield, C. Wesdemiotis, S. Z. Cheng, G. R. Newkome, *Angew. Chem. Int. Ed.* **2013**, 52, 7728.
- [23] a) M. Wang, C. Wang, X.-Q. Hao, J. Liu, X. Li, C. Xu, A. Lopez, L. Sun, M.-P. Song, H.-B. Yang, X. Li, *J. Am. Chem. Soc.* **2014**, 136, 6664; b) H. Wang, X. Qian, K. Wang, M. Su, W.-W. Haoyang, X. Jiang, R. Brzozowski, M. Wang, X. Gao, Y. Li, B. Xu, P. Eswara, X.-Q. Hao, W. Gong, J.-L. Hou, J. Cai, X. Li, *Nat. Comm.* **2018**, 9, 1815; c) G.-Q. Yin, H. Wang, X.-Q. Wang, B. Song, L.-J. Chen, L. Wang, X.-Q. Hao, H.-B. Yang, X. Li, *Nat. Comm.* **2018**, 9, 567.
- [24] a) T.-Z. Xie, K. J. Endres, Z. Guo, J. M. Ludlow, C. N. Moorefield, M. J. Saunders, C. Wesdemiotis, G. R. Newkome, *J. Am. Chem. Soc.* **2016**, 138, 12344; b) S. Chakraborty, W. Hong, K. J. Endres, T.-Z. Xie, L. Wojtas, C. N. Moorefield, C. Wesdemiotis, G. R. Newkome, *J. Am. Chem. Soc.* **2017**, 139, 3012.
- [25] a) M. Wang, C. Wang, X.-Q. Hao, X. Li, T. J. Vaughn, Y.-Y. Zhang, Y. Yu, Z.-Y. Li, M.-P. Song, H.-B. Yang, X. Li, *J. Am. Chem. Soc.* **2014**, 136, 10499; b) M. Chen, J. Wang, D. Liu, Z. Jiang, Q. Liu, T. Wu, H. Liu, W. Yu, J. Yan, P. Wang, *J. Am. Chem. Soc.* **2018**, 140, 2555.
- [26] Z. Zhang, H. Wang, X. Wang, Y. Li, B. Song, O. Bolarinwa, R. A. Reese, T. Zhang, X.-Q. Wang, J. Cai, B. Xu, M. Wang, C. Liu, H.-B. Yang, X. Li, *J. Am. Chem. Soc.* **2017**, 139, 8174.
- [27] S. Leininger, J. Fan, M. Schmitz, P. J. Stang, *Proc. Natl. Acad. Sci.* **2000**, 97, 1380.
- [28] a) K. C. D. Robson, B. D. Koivisto, T. J. Gordon, T. Baumgartner, C. P. Berlinguette, *Inorg. Chem.* **2010**, 49, 5335; b) S. Baitalik, X.-Y. Wang, R. H. Schmehl, *J. Am. Chem. Soc.* **2004**, 126, 16304.